Cultivator



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Cultivator

What it is used for

A cultivator loosens and aerates soil, controls weeds, and prepares the seedbed using tractor-mounted, powered, or manually operated tines. Correct adjustment, suitable soil conditions, and safe operation help improve field preparation while reducing equipment damage and operator injury.

Main parts

- Frame supports the cultivator assembly.
- Tines or shanks loosen soil and disturb weeds.
- Shares or sweeps cut and lift soil where fitted.
- Hitch or mounting points connect the cultivator to a tractor where applicable.
- Depth adjustment controls working depth where provided.
- Drive or power transmission components operate powered models.
- Wheels or skids support the machine and help control depth.
- Guards and shields protect the operator from moving components where fitted.

Before you start

1. Only trained operators should use tractor-mounted or powered cultivators.
2. Read the operator manual and confirm the cultivator is suitable for the tractor or power source.
3. Inspect the frame, tines, shares, fasteners, hitch, wheels, guards, and drive components for damage.
4. For tractor-mounted equipment, park on level ground, apply the parking brake, lower attachments, and shut off the engine before connecting or adjusting.
5. Check the field for stones, wire, buried services, people, and other obstructions.
6. Wear fitted work clothing, safety footwear, eye protection where needed, and hearing protection when required.

How to use

1. Connect and secure the cultivator using the approved hitching method where applicable.
2. Set the working depth and tine arrangement according to the crop, soil, and field conditions.
3. Confirm guards are fitted and no tools or loose items remain near moving parts.
4. Start the tractor or powered unit from the operator position and check for abnormal noise or vibration.
5. Begin at a low, controlled speed and engage the cultivator according to the manual.

6. Maintain a steady path and adjust speed or depth if soil conditions cause excessive bouncing or blockage.
7. Raise or disengage the cultivator before turning or travelling over unsuitable ground as directed by the manual.
8. Stop the machine, isolate power, and wait for all movement to cease before clearing blockages or making adjustments.
9. After work, lower the attachment safely, disconnect power where applicable, and inspect the tines and frame.
10. Clean off soil and plant residue, record faults, and park the machine securely.

⚠ Safety

- Never approach rotating or moving tines while the machine is running.
- Shut off the engine or power source, remove the key where applicable, and wait for all parts to stop before clearing a blockage.
- Never carry passengers on the cultivator or allow bystanders near the working area.
- Keep hands and feet away from pinch points during hitching and adjustment; use approved lifting supports where needed.
- Do not operate on unstable slopes or near ditches beyond the machine's approved limits.
- Use guards and follow tractor PTO and power-transmission safety instructions where relevant.

Care & storage

- Remove soil and plant residue after use, especially around tines and moving parts.
- Inspect tines, shares, fasteners, hitch points, wheels, and guards before and after operation.
- Lubricate designated points and tighten fasteners according to the maintenance schedule.
- Replace worn or damaged parts with compatible components and arrange authorised repairs when required.

Common mistakes to avoid

- Using an unsuitable working depth or speed for the soil conditions.
- Operating with loose tines, damaged guards, or an insecure hitch.
- Clearing blockages while the engine or power source is running.
- Failing to inspect the field for people, stones, wire, or hidden obstructions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.